

A New Table of Abundances of the Elements in the Solar System

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Abstract

A new table of the abundances of the elements, presumably characteristic of primitive solar matter, has been compiled. It is based as much as possible on abundances in Type I carbonaceous chondrites. Other sources, used where necessary, are ordinary chondrites, solar atmospheric abundances, and solar cosmic ray abundances. Eight elements were interpolated using criteria based on the theory of nucleosynthesis in stars. A discussion is given of some features of the abundance table which should be taken into account in theories of nucleosynthetic processes.

A basic and necessary step in the development of theories of stellar nucleosynthesis was made by Suess and Urey (1956), who selected astronomical, meteoritic, and terrestrial element abundance data and produced a table of cosmic abundances. Most of the abundance data used by Suess and Urey were of poor quality, so these authors adjusted and interpolated the abundances of the elements to produce a smooth variation in the abundances of nuclides with odd mass numbers. From the resulting abundance tables it was then possible to identify a number of different nuclear processes which could probably occur in stars and which would produce the characteristic features of the Suess-Urey table (Burbidge, Burbidge, Fowler and Hoyle, 1957; Cameron, 1957).

The writer (Cameron, 1959) later readjusted some parts of the Suess-Urey abundance table making use of the hindsight given by the identification of the processes responsible for nucleosynthesis to use additional criteria for the adjustment of the abundances. Apart from minor adjustments, the principal predictions were that the Sr/Zr ratio should be inverted, the rare earth abundances should be strongly modified, and the abundance of Pb should be greatly increased. Subsequent meteoritic abundance measurements have confirmed

the first two of these points and have shown that the Pb abundance lies between the two values suggested.

In subsequent years a great deal of high-quality meteoritic abundance data has been obtained by radiochemical techniques, and interest has arisen in the differences in abundances between various meteorite classes. The writer (Cameron, 1963) made a new abundance compilation based to a much greater extent than previously on good meteorite abundance measurements. Although this table was not published, many references to it appear in the literature, and hence the numbers obtained at that time have been included in Table 1 for the sake of the record.

Anders (1964) has pointed out that there are systematic differences between the abundances of certain elements of somewhat greater volatility between abundance classes; he postulated some mechanisms to account for this. These ideas have recently received additional discussion (Larimer and Anders, 1967). These authors and others have pointed out that the least element depletions have occurred in Type I carbonaceous chondrites. Taking the point of view that it is probably easier to obtain uniform element depletion factors than uniform enrichment factors, the present writer has constructed a new abundance table based as much as possible on measurements in Type I carbonaceous chondrites. These meteorites are thus assumed to be the most representative source of solar system abundance data for which high-quality measurements can be made.

The resulting list of abundances is given in Table 1 as Cameron (1967). The sources of the abundances and the assumptions made in assigning values to certain elements are given in the notes to Table 1. It may be seen from these notes that the majority of the nonvolatile elements now have abundances based on Type I carbonaceous chondrite measurements. In some cases it has been necessary to use abundances measured in ordinary chondrites. The volatile elements are based on solar photospheric and cosmic ray abundance data; they have been normalized to the meteoritic data using the solar photospheric abundances of ten nonvolatile elements. There remains a significant uncertainty in this normalization factor, perhaps amounting to a few tens of percent. Only eight elements have now been interpolated on the basis of nuclear regularities.

It should be noted that this procedure has selected a high value of the iron abundance. This abundance has been a long-standing problem, since the solar photospheric abundance of iron (Aller, 1961) is considerably lower than the meteoritic value, and the oscillator strengths of the iron lines used in the solar abundance determination are claimed to be of superior quality. These are strong iron lines. Nevertheless, it appears that the iron abundance determined from weak lines is higher than for the strong lines (L. Goldberg, private communication), and the iron abundance is anomalously high in the solar corona. Hence it has seemed best to use the meteoritic value. The question of

TABLE 1. COMPILATIONS OF ABUNDANCES NORMALIZED TO Si = 10^6
(Notes refer to the sources of Cameron (1967))

Element	Suess Urey	Cameron (1963)	Cameron (1967)	Notes
1 H	4.00×10^{10}	3.2×10^{10}	2.6×10^{10}	1
2 He	3.08×10^9	5.0×10^9	2.1×10^9	2
3 Li	100	38	45	3
4 Be	20	7	0.69	3
5 B	24	6	6.2	3
6 C	3.5×10^6	1.66×10^7	1.35×10^7	1
7 N	6.6×10^6	3.0×10^6	2.44×10^6	1
8 O	2.15×10^7	2.9×10^7	2.36×10^7	1
9 F	1600	$\sim 10^4$	3630	4
10 Ne	8.6×10^6	1.7×10^7	2.36×10^6	2
11 Na	4.38×10^4	4.18×10^4	6.32×10^4	4
12 Mg	9.12×10^5	1.046×10^6	1.050×10^6	5
13 Al	9.48×10^4	8.93×10^4	8.51×10^4	5
14 Si	1.00×10^6	1.00×10^6	1.00×10^6	5
15 P	1.00×10^4	9320	1.27×10^4	5
16 S	3.75×10^5	6.0×10^5	5.06×10^5	4
17 Cl	8850	1836	1970	4
18 Ar	1.4×10^5	2.4×10^5	2.28×10^5	6
19 K	3160	2970	3240	4
20 Ca	4.90×10^4	7.28×10^4	7.36×10^4	5
21 Sc	28	29	33	5
22 Ti	2240	3140	2300	5
23 V	220	590	900	7
24 Cr	7800	1.20×10^4	1.24×10^4	5
25 Mn	6850	6320	8800	4
26 Fe	6.00×10^5	8.42×10^5	8.90×10^5	5
27 Co	1800	2290	2300	5
28 Ni	2.74×10^4	4.44×10^4	4.57×10^4	5
29 Cu	212	861	919	4
30 Zn	486	930	1500	4
31 Ga	11.4	39	45.5	4
32 Ge	50.4	134	126	4
33 As	4.0	4.4	7.2	8
34 Se	67.6	18.8	70.1	4
35 Br	13.4	3.95	20.6	4
36 Kr	51.3	20	64.4	9
37 Rb	6.5	5.0	5.95	4
38 Sr	18.9	21	58.4	5
39 Y	8.9	3.6	4.6	5
40 Zr	54.5	23	30	7
41 Nb	1.00	0.81	1.15	10
42 Mo	2.42	2.42	2.52	3
44 Ru	1.49	1.58	1.6	3
45 Rh	0.214	0.26	0.33	3
46 Pd	0.675	1.00	1.5	11
47 Ag	0.26	0.26	0.5	12
48 Cd	0.89	0.89	2.12	4
49 In	0.11	0.11	0.217	4
50 Sn	1.33	1.33	4.22	4
51 Sb	0.246	0.15	0.381	4

TABLE 1.—*continued*

Element	Suess Urey	Cameron (1963)	Cameron (1967)	Notes
52 Te	4.67	3.00	6.76	4
53 I	0.80	0.46	1.41	4
54 Xe	4.0	3.15	7.10	13
55 Cs	0.456	0.25	0.367	4
56 Ba	3.66	4.0	4.7	5
57 La	2.00	0.38	0.36	5
58 Ce	2.26	1.08	1.17	5
59 Pr	0.40	0.16	0.17	5
60 Nd	1.44	0.69	0.77	5
62 Sm	0.664	0.24	0.23	5
63 Eu	0.187	0.083	0.091	5
64 Gd	0.684	0.33	0.34	14
65 Tb	0.0956	0.054	0.052	14
66 Dy	0.556	0.33	0.36	5
67 Ho	0.118	0.076	0.090	5
68 Er	0.316	0.21	0.22	5
69 Tm	0.0318	0.032	0.035	5
70 Yb	0.220	0.18	0.21	5
71 Lu	0.050	0.031	0.035	5
72 Hf	0.438	0.16	0.16	3
73 Ta	0.065	0.021	0.022	15
74 W	0.49	0.11	0.16	11
75 Re	0.135	0.054	0.055	16
76 Os	1.00	0.73	0.71	16
77 Ir	0.821	0.500	0.43	17
78 Pt	1.625	1.157	1.13	18
79 Au	0.145	0.13	0.20	17
80 Hg	0.284	0.27	0.75	19
81 Tl	0.108	0.11	0.182	4
82 Pb	0.47	2.2	2.90	4
83 Bi	0.144	0.14	0.164	4
90 Th	—	0.069	0.034	20
92 U	—	0.042	0.0234	20

Notes to Table 1

1. In Cameron (1963) the volatile elements are normalized to silicon = 10^6 in the sun, which is somewhat poorly determined. In Cameron (1967) the normalization is based on logarithmic averages of Na, Mg, Al, Si, S, K, Ca, Ti, Co, and Ni in the sun relative to meteoritic values. The solar abundances are taken from Ailer (1961).

2. He and Ne are normalized to the solar oxygen abundance using solar cosmic ray data, as suggested by Gaustad (1964).

3. Based on abundances in chondrites, taken from Urey (1964).

4. Based on abundances in Type I carbonaceous chondrites, taken from Larimer and Anders (1967).

5. Based on abundances in the Orgueil and Ivuna Type I carbonaceous chondrites, taken from Urey (1964).

6. Ar^{36} was interpolated between S^{32} and Ca^{40} .

7. Based on abundances in carbonaceous chondrites, taken from Urey (1964).

8. As^{75} was interpolated between Ge^{73} and Se^{77} .

the iron abundance is of great importance for problems of nucleosynthesis, and it is important that additional work be done in an attempt to resolve the problem.

The abundances of individual nuclides corresponding to the latest adjustment are given in Table 2. Also given in this table is a classification of certain of the heavier isobars as being predominantly made by a particular mechanism of nucleosynthesis. Thus F isobars are those which have been made principally by neutron capture on a fast time scale, corresponding to the *r*-process of Burbidge *et al.* (1957). S isobars are those which have been made principally by neutron capture on a slow time scale, corresponding to the *s*-process of Burbidge *et al.* (1957). B isobars are those which are bypassed by neutron capture processes, corresponding to the *p*-process of Burbidge *et al.* (1957).

A few assumptions have been necessary for the construction of Table 2 in addition to those used in the construction of Table 1. Thus the He^3/He^4 ratio has been taken to be 3×10^{-4} since this ratio is usually found in primordial helium extracted from meteorites (Signer and Suess, 1963). The large abundance of Ar^{40} in atmospheric argon is due to decay of terrestrial

Notes to Table 1.—continued.

9. Kr was assigned an abundance representing a compromise between interpolations of Kr^{84} between Se^{80} and Sr^{88} and of Kr^{83} between Br^{81} and Rb^{85} .

10. Nb^{93} was interpolated between Zr^{91} and Mo^{95} .

11. Based on abundances in Type I carbonaceous chondrites, private communication from E. Anders.

12. Based on abundances in carbonaceous chondrites, taken from Larimer and Anders (1967).

13. Isotopic abundances of Xe were based on those in the Murray carbonaceous chondrite (Cameron, 1962). The element was normalized to Te and I to form a continuous abundance peak.

14. The abundances of Gd and Tb measured for Type I carbonaceous chondrites are anomalous relative to the usual chondritic pattern and have fairly large errors. Hence these abundances were based on those in ordinary chondrites (Urey, 1964) but normalized to Sm.

15. The Ta abundance of 0.044 for carbonaceous chondrites (Urey, 1964) appears too high relative to neighboring odd-mass nuclides. The given value was interpolated between Hf^{179} and W^{183} and is in good agreement with values in ordinary chondrites.

16. Based on abundances in Type I carbonaceous chondrites, taken from Morgan and Lovering (1967).

17. Based on abundances in Type I carbonaceous chondrites, taken from Baedecker (1967).

18. The abundance of Pt was adjusted relative to Os, Ir, and Au to form a continuous peak. The adopted value is slightly higher than in ordinary chondrites (Urey, 1964).

19. The abundances of Hg in carbonaceous chondrites are anomalously high and greatly variable. Hg was interpolated with as high a value as seemed possible, specifically by interpolating Hg^{199} and Hg^{201} between Au^{197} and Tl^{203} .

20. Abundances as of 4.5×10^9 years ago based on ordinary chondrites, taken from Urey (1964).

TABLE 2. ABUNDANCES OF THE NUCLIDES CORRESPONDING TO THE
ELEMENT ABUNDANCES OF CAMERON (1967) IN TABLE 1

Element	A	% Abundance	Class	Abundance	
H	1	99.985		2.6×10^{10}	2.6×10^{10}
	2	0.015			3.9×10^6
He	3	~ 0.03		2.1×10^9	6.3×10^5
	4	~ 100			2.1×10^9
Li	6	7.42		45	3.34
	7	92.58			41.66
Be	9	100		0.69	
B	10	19.64		6.2	1.22
	11	80.36			4.98
C	12	98.89		1.35×10^7	1.33×10^7
	13	1.11			1.50×10^5
N	14	99.634		2.44×10^6	2.44×10^6
	15	0.366			9000
O	16	99.759		2.36×10^7	2.36×10^7
	17	0.0374			8800
F	18	0.2039			4.8×10^4
	19	100		3630	
Ne	20	90.92		2.36×10^6	2.15×10^6
	21	0.257			6060
Na	22	8.82			2.00×10^5
	23	100		6.32×10^4	
Mg	24	78.70		1.050×10^6	8.26×10^5
	25	10.13			1.06×10^5
Al	26	11.17			1.17×10^5
	27	100		8.51×10^4	
Si	28	92.21		1.000×10^6	9.22×10^5
	29	4.70			4.70×10^4
P	30	3.09			3.09×10^4
	31	100		1.27×10^4	
S	32	95.0		5.06×10^5	4.80×10^5
	33	0.760			3.85×10^3
Cl	34	4.22	F		2.14×10^4
	36	0.0136			69
Ar	35	75.529		1970	1490
	37	24.471			480
	36	84.2		2.28×10^5	1.92×10^5
	38	15.8			3.61×10^4
	40	~ 0.01	F		~ 20

TABLE 2.—continued

Element	A	% Abundance	Class	Abundance	
K				3240	
	39	93.10			3020
	40	0.38			12.3
Ca	41	6.88			223
				7.36×10^4	
	40	96.97			7.13×10^4
	42	0.64			470
	43	0.145			107
	44	2.06			1520
	46	0.0033	F		2.43
Sc	48	0.185	F		136
	45	100		33	
Ti				2300	
	46	7.93			182
	47	7.28			168
	48	73.94			1700
	49	5.51			127
	50	5.34			123
V				900	
	50	0.24			2.2
Cr	51	99.76			898
				1.24×10^4	
	50	4.31			535
	52	83.7			1.04×10^4
	53	9.55			1180
Mn	54	2.38			297
	55	100		8800	
Fe				8.90×10^5	
	54	5.82			5.19×10^4
	56	91.66			8.15×10^5
	57	2.19			1.95×10^4
	58	0.33			2900
Co	59	100		2300	
				4.57×10^4	
Ni	58	67.88			3.10×10^4
	60	26.23			1.20×10^4
	61	1.19			544
	62	3.66			1670
	64	1.08			494
Cu				919	
	63	69.09			635
Zn	65	30.91			284
				1500	
	64	48.89			732
	66	27.81			417
	67	4.11			61.8
	68	18.57			278
	70	0.62	F		9.3
Ga				45.5	
	69	60.4			27.5
	71	39.6			18.0

TABLE 2.—*continued*

Element	A	% Abundance	Class	Abundance	
Ge				126	
	70	20.52	S		25.8
	72	27.43			34.6
	73	7.76			9.77
	74	36.54			46.0
	76	7.76	F		9.77
As	75	100		7.2	
Se				70.1	
	74	0.87	B		0.61
	76	9.02	S		6.32
	77	7.58			5.31
	78	23.52			16.5
	80	49.82			34.9
	82	9.19	F		6.44
Br				20.6	
	79	50.537			10.4
Kr	81	49.463			10.2
				64.4	
	78	0.354	B		0.228
	80	227			1.46
	82	11.56	S		7.45
	83	11.55			7.44
	84	56.90			36.6
	86	17.37			11.2
Rb				5.95	
	85	72.15			4.29
Sr	87	27.85			1.66
				58.4	
	84	0.56	B		0.327
	86	9.86			5.76
	87	7.02			4.10
	88	82.56			48.2
Y	89	100		4.6	
Zr				30	
	90	51.46			15.1
	91	11.23			3.37
	92	17.11			5.14
	94	17.40			5.32
	96	2.80	F		0.84
Nb	93	100		1.15	
Mo				2.52	
	92	15.84	B		0.399
	94	9.04	B		0.228
	95	15.72			0.396
	96	16.53	S		0.417
	97	9.46			0.238
	98	23.78			0.599
	100	9.63	F		0.242
Ru				1.6	
	96	5.51	B		0.088
	98	1.87	B		0.030
	99	12.72			0.204

TABLE 2.—continued

Element	A	% Abundance	Class	Abundance
Rh	100	12.62	S	0.202
	101	17.07		0.273
	102	31.61		0.505
	104	18.58	F	0.297
	103	100		0.33
Pd				1.5
	102	0.96	B	0.0144
	104	10.97	S	0.164
	105	22.23		0.324
	106	27.33		0.410
Ag	108	26.71		0.400
	110	11.81	F	0.177
				0.5
	107	51.35		0.257
	109	48.65		0.243
Cd				2.12
	106	1.215	B	0.0258
	108	0.875	B	0.0186
	110	12.39	S	0.262
	111	12.75		0.270
In	112	24.07		0.510
	113	12.26		0.260
	114	28.86		0.611
	116	7.58	F	0.161
				0.217
Sn	113	4.28		0.0093
	115	95.72		0.208
				4.22
	112	0.96	B	0.0405
	114	0.66	(B)	0.0278
Sb	115	0.35	(B)	0.0148
	116	14.30	S	0.603
	117	7.61		0.321
	118	24.03		1.01
	119	8.58		0.362
Te	120	32.85		1.39
	122	4.72	(F)	0.199
	124	5.94	F	0.250
				0.381
	121	57.25		0.218
I	123	42.75		0.163
				6.76
	120	0.089	B	0.0060
	122	2.46	S	0.166
	123	0.87		0.059
	124	4.61	S	0.312
	125	6.99		0.472
	126	18.71		1.27
	128	31.79	F	2.15
	130	34.48	F	2.33
				1.41
	127	100		

TABLE 2.—*continued*

Element	A	% Abundance	Class	Abundance	
Xe				7.10	
	124	0.126	B		0.00895
	126	0.115	B		0.00816
	128	2.17	S		0.154
	129	27.5			1.95
	130	4.26	S		0.303
	131	21.4			1.52
	132	26.0			1.85
	134	10.17	F		0.721
	136	8.39	F		0.595
Cs	133	100		0.367	
Ba				4.7	
	130	0.101	B		0.00475
	132	0.097	B		0.00455
	134	2.42	S		0.114
	135	6.59			0.310
	136	7.81	S		0.367
	137	11.32			0.532
	138	71.66			3.37
La				0.36	
	138	0.089			0.00032
	139	99.911			0.36
Ce				1.17	
	136	0.193	B		0.00226
	138	0.250	B		0.00292
	140	88.48			1.035
Pr	142	11.07	F		0.129
	141	100		0.17	
				0.77	
Nd	142	27.11	S		0.209
	143	12.17			0.0936
	144	23.85			0.184
	145	8.30			0.0640
	146	17.22			0.133
	148	5.73	F		0.0441
	150	5.62	F		0.0433
Sm				0.23	
	144	3.09	B		0.0071
	147	14.97			0.0344
	148	11.24	S		0.0259
	149	13.83			0.0318
	150	7.44	S		0.0171
	152	26.72			0.0614
Eu	154	22.71	F		0.0522
				0.091	
	151	47.82			0.0435
Gd	153	52.18			0.0475
				0.34	
	152	0.200			0.00068
	154	2.15	S		0.00731
	155	14.73			0.0501
	156	20.47			0.0696

TABLE 2.—*continued*

Element	A	% Abundance	Class	Abundance
Tb	157	15.68	F	0.0532
	158	24.87		0.0847
	160	21.90		0.0745
	159	100		0.052
Dy				0.36
	156	0.0524	B	0.000189
	158	0.0902	B	0.000325
	160	2.294	S	0.00825
	161	18.88		0.0680
	162	25.53		0.0919
	163	24.97		0.0899
	164	28.18		0.1112
Ho	165	100		0.090
Er				0.22
	162	0.136	B	0.000299
	164	1.56		0.00343
	166	33.41		0.0735
	167	22.94		0.0505
	168	27.07		0.0595
	170	14.88	F	0.0327
Tm	169	100		0.035
Yb				0.21
	168	0.135	B	0.000284
	170	3.03	S	0.00636
	171	14.31		0.0301
	172	21.82		0.0460
	173	16.13		0.0339
	174	31.84		0.0670
	176	12.73	F	0.0268
Lu				0.035
	175	97.41		0.0341
	176	2.59		0.00091
Hf				0.16
	174	0.18	B	0.000288
	176	5.20	S	0.00832
	177	18.50		0.0296
	178	27.14		0.0434
	179	13.75		0.0220
	180	35.24		0.0564
Ta				0.022
	180	0.0123		0.0000027
	181	99.9877		0.022
W				0.16
	180	0.135	B	0.000216
	182	26.41		0.0423
	183	14.40		0.0230
	184	30.64		0.0490
	186	28.41	F	0.0455
Re				0.055
	185	37.07		0.0204
	187	62.93		0.0346

TABLE 2.—*continued*

Element	A	% Abundance	Class	Abundance	
Os				0.71	
	184	0.018	B		0.000128
	186	1.29	S		0.0113
	187	1.64			0.0116
	188	13.3			0.0945
	189	16.1			0.1142
	190	26.4			0.1875
	192	41.0	F		0.291
Ir				0.43	
	191	37.3			0.164
	193	62.7			0.276
Pt				1.13	
	190	0.0127	B		0.000144
	192	0.78	S		0.00994
	194	32.9			0.372
	195	33.8			0.382
	196	25.3			0.286
	198	7.21	F		0.0815
	197	100		0.20	
Au				0.75	
Hg					
	196	0.146	B		0.00102
	198	10.02	S		0.0703
	199	16.84			0.118
	200	23.13			0.162
	201	13.22			0.0926
	202	29.80			0.2084
	204	6.85	F		0.0480
Tl				0.182	
	203	29.50			0.0537
	205	70.50			0.1283
Pb				2.90	
	204	1.97	S		0.057
	206	18.83			0.546
	207	20.60			0.596
	208	58.55			1.70
Bi	209	100		0.164	
Th	232	100		0.034	
U				0.0234	
	235	23.1			0.0054
	238	76.9			0.0180

K^{40} ; the estimate in Table 2 comes from the consideration that Ar^{40} is a product of neutron capture on a fast time scale and hence should be comparable in abundance to S^{36} , Ca^{46} , and Ca^{48} . The abundances of uranium, thorium, and K^{40} isotopes have been corrected for a decay interval of 4.5×10^9 years to make them typical of the initial solar system; other radioactive nuclides have not been corrected because the half-lives are long and the corrections small. The isotopic abundances of xenon have been taken from the analysis of xenon from the Murray carbonaceous chondrite according to

the interpretation given by Cameron (1962); this assumes that the atmospheric composition of xenon is distorted by a fission xenon contribution.

The Cameron (1967) abundances are plotted as a function of mass number in figure 1. Of particular interest for problems of nucleo-synthesis are the odd-even ratios among the heavier nuclei and the sharp and rounded peaks corresponding to closed shells of 82 and 126 neutrons in the process of neutron capture on slow and fast time scales.

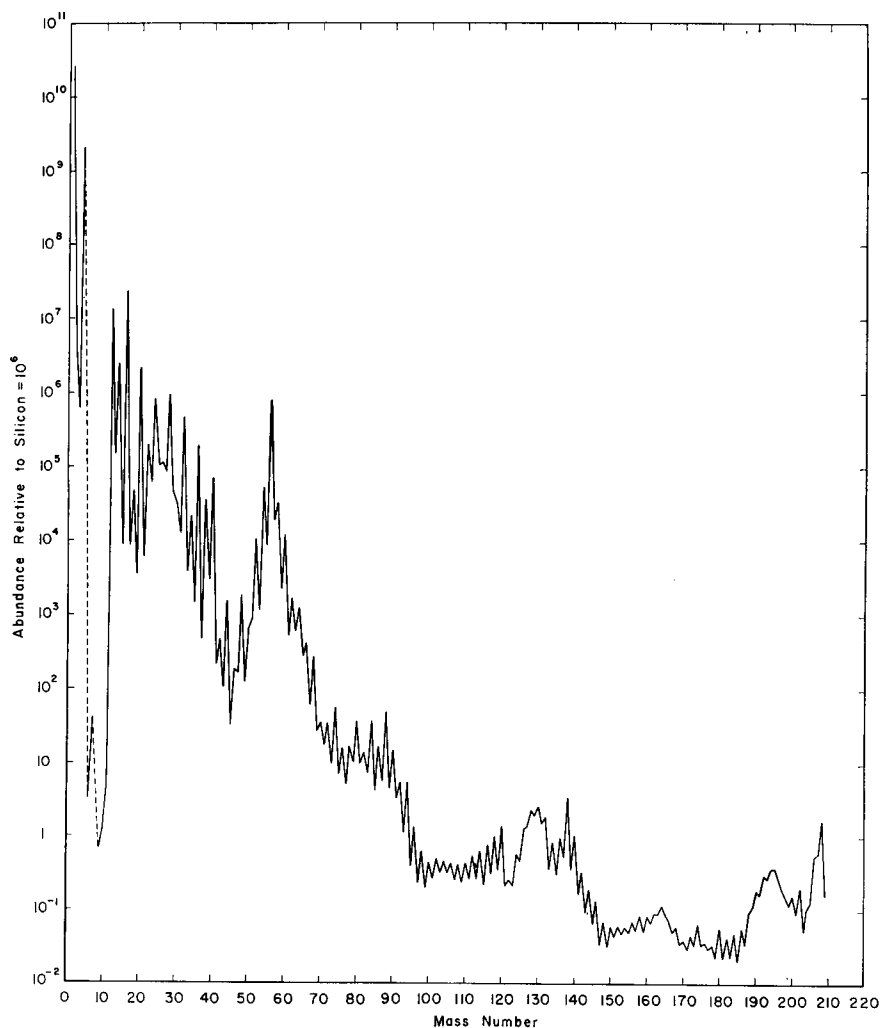


FIG. 1. Abundances of the nuclides plotted as a function of mass number.

The abundances of products of neutron capture are shown separately for each element upwards from the iron region in figures 2 to 5. Nuclides with odd mass numbers are plotted as solid circles; those with even mass numbers are plotted with crosses. If an even mass number is classed as an F isobar, the cross is surrounded by a square; if it is classed as an S isobar, the cross is

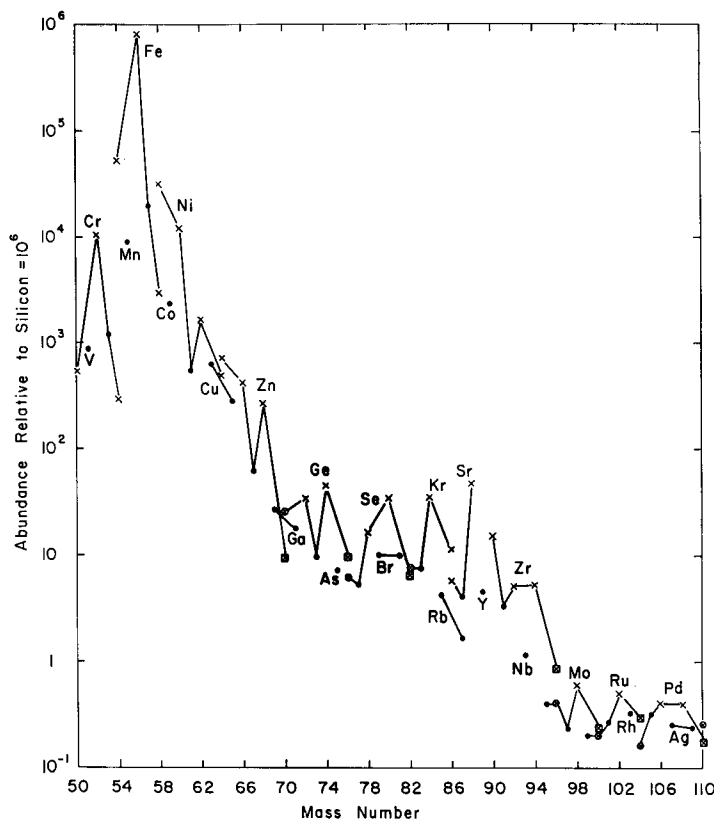


FIG. 2. Abundances of nuclides which can be formed by neutron capture, upwards from the iron peak region (in which all nuclides are plotted). Symbols are explained in the text.

surrounded by a circle. These figures are very useful if one wishes to make estimates of the relative contributions to abundances of nuclides from the fast and slow time scale processes.

It should be emphasized that the abundances of heavy elements which are now based principally upon good determinations in Type I carbonaceous chondrites are very satisfactory from the point of view of the processes of nucleosynthesis. It has been known for a long time that there is an excellent

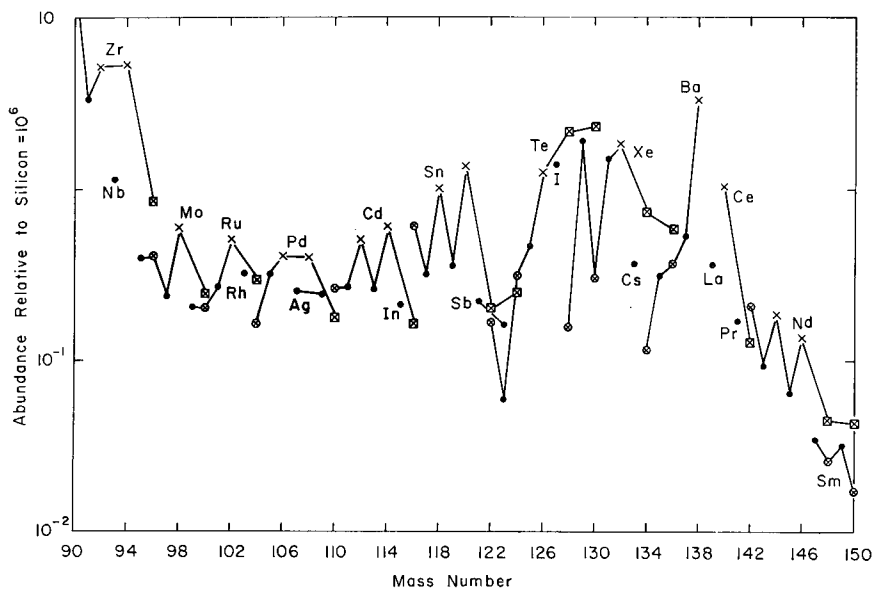


FIG. 3. Continuation of figure 2.

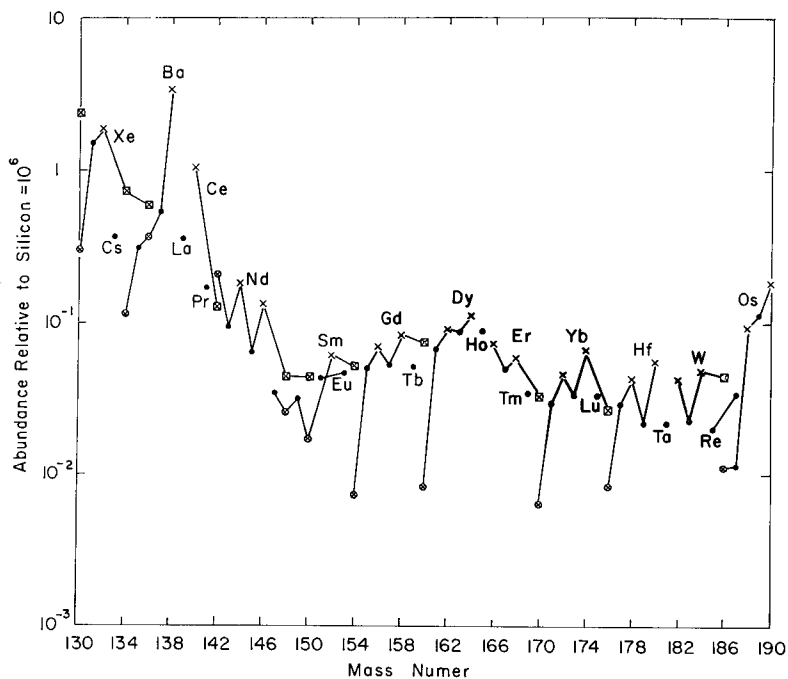


FIG. 4. Continuation of figure 3.

regularity in abundances of nuclides corresponding to a given process of nucleosynthesis among the rare earth elements, which are subject to very little chemical fractionation. This property is now generally true of the entire distributions of figures 2 to 5. Odd-even ratios vary regularly, being small for fast time scale products and larger for slow time scale products.

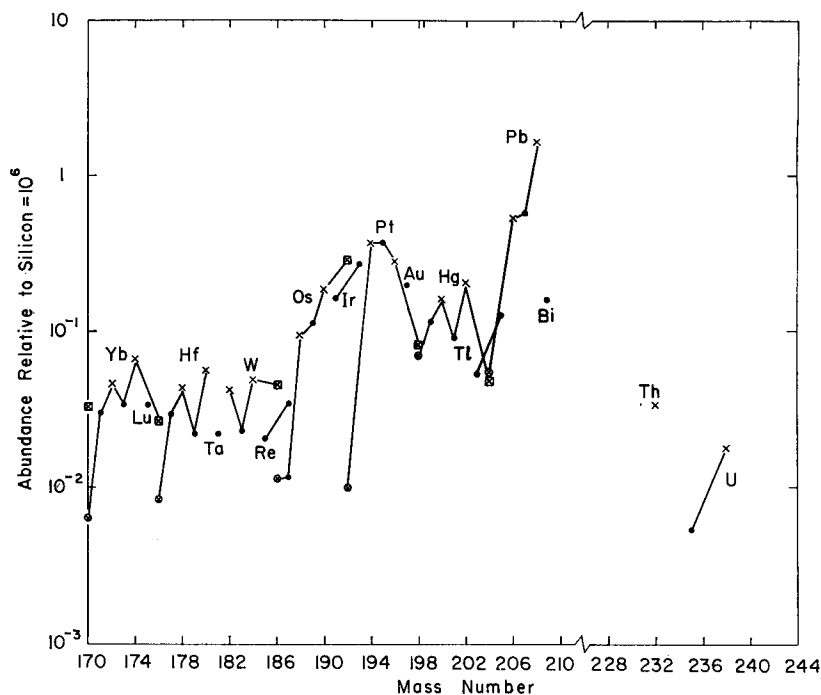


FIG. 5. Continuation of figure 4.

One difficulty with previous abundance determinations has been the small odd-even ratios for the isotopes of Cu and Zn. This has now been satisfactorily eliminated because of the higher Zn abundance measured in Type I carbonaceous chondrites. Other expected regularities as enumerated by Cameron (1959) are satisfactorily followed.

In figure 6 are plotted the abundance trends of even mass numbers for three processes of nucleosynthesis: products of neutron capture on fast and slow time scales, and bypassed nuclei. These abundance trends have been determined from isobaric abundances and from additional behavior, such as peaks at closed neutron shells, where it is evident that the observed abundances are produced primarily by a single process.

One of the striking features of the abundance distribution of the products of neutron capture on a slow time scale is the fact that σN , the product of the

average neutron capture cross-section near 25 keV and the abundance, is a smooth monotonically decreasing function of mass number (Seeger, Fowler and Clayton, 1965). Because of local variability in the neutron capture cross-sections, it may be noticed in figure 6 that the abundances of the S isobars scatter about a smooth curve.

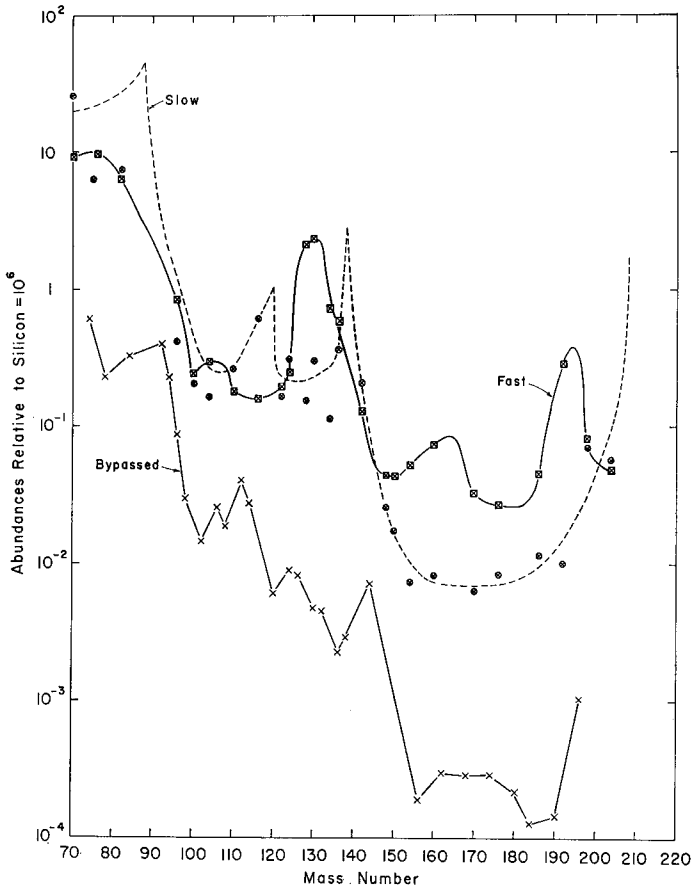


FIG. 6. Trends among heavy nuclei of abundances attributable to neutron capture on fast and slow time scales and to bypassed processes.

At present we know much less about the set of astrophysical environments responsible for the production of the products of neutron capture on a fast time scale than we do about those responsible for the slow time scale products. Therefore it is important to examine the abundance data to determine whether additional boundary conditions can be assigned to the fast time scale processes. Several features are worthy of note. The curve drawn through

the F isobar abundances is remarkably smooth, in contrast to the scatter about the S isobar curve. Hence the fast time scale abundances are not affected by the variability of individual neutron capture cross-sections. Abundance smoothing processes must have been operative, either through a contribution from several values of atomic number Z to each mass number, or through frequent neutron emission following the high-energy beta decays of the neutron-rich final products of the capture process. In addition, the neutron capture process must have terminated quite abruptly; otherwise fast beta decays would have produced only one capture product per mass number, and the final abundances would have been affected by cross-section variations.

The fast time scale product peaks at mass numbers 130 and 195 correspond to closed shells of 82 and 126 neutrons. As discussed by Truran, Arnett, Tsuruta and Cameron (1967), it appears that these peaks are made near the base of an ejected supernova envelope, where the material has been largely transformed into neutrons. Under the astrophysical conditions in which these peaks are made, it is expected that negligible amounts of intermediate mass numbers will be formed. Yet it can be seen in figure 6 and from Table 2 that the abundances associated with F isobars near mass numbers 40 and 70 are much larger than those in the closed shell peaks mentioned above. This suggests that there must be a very different fast time scale process in which a burst of neutrons can be produced and added to pre-existing intermediate elements. One possibility for producing this situation is the passage of a supernova shock wave through the helium shell in the presupernova structure, where rapid (α, n) reactions may be an effective neutron source.

Also plotted in figure 6 are the abundances of the B isobars. These can be produced from a pre-existing abundance distribution either by photo-disintegrations which remove neutrons and protons from nuclei, or by proton capture reactions taking place in a hydrogen-rich region briefly raised to a high temperature. The first case can arise when a supernova shock wave traverses a hydrogen-depleted region of the presupernova structure in which neutrons cannot be produced in quantity by (α, n) reactions. The second case can occur when a supernova shock wave traverses the outer hydrogen layer of the presupernova structure.

The abundance distribution of figure 6 favors the proton capture process. Since this would take place in the outer layers of the presupernova structure, the heavy elements could at most have been exposed to the very small neutron flux accompanying deuterium-burning, and the abundance changes produced by this would be negligible. Hence one should expect that the abundance distribution of the proton capture products would be similar to the superposition of the abundance distributions produced by neutron capture, except that the increasing Coulomb barrier with increasing atomic number would produce a progressive decrease in proton capture rates, with a corresponding increase in the ratio of the abundances of the neutron capture to proton

capture products. It may be seen that this expectation is fulfilled. In addition, the abundances of the B isobars are relatively high up to a position slightly beyond the slow time scale closed shell peaks, beyond which they fall rapidly. This also would be expected on the proton capture mechanism.

The rise at the upper end of the B isobar distribution is given solely by an isotope of mercury. This element was interpolated in the compilation of the new abundance table, and there is great uncertainty in the interpolation. The mercury abundance was chosen as high as seemed reasonable in view of the large amounts of mercury in carbonaceous chondrites. Hence the reality of this final rise in the B isobar distribution is not at all certain.

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